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An Explainable and Adaptive Zero-Day Attack Detection Framework for Internet of Medical Things (IoMT) Environments Using Self-Supervised Learning and Graph-Based Anomaly Detection[cite: 1]

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ABSTRACT

The Internet of Medical Things (IoMT) improves healthcare by interconnecting medical devices but presents cybersecurity concerns, particularly zero-day attacks affecting patient safety and data privacy. Traditional Intrusion Detection Systems (IDS) cannot detect unknown threats. AI based approaches suffer from generalization, false alarms, explain ability and efficiency issues. This research proposes an Explainable and Adaptive Zero-Day Attack Detection Framework based on the fusion of self-supervised learning, graph-based anomaly detection and Explainable Artificial Intelligence (XAI). The framework learns normal communication patterns and device relationships and uses this information to detect previously unseen attacks without requiring fully labeled data. The key performance indicators include detection rate, precision, recall, F1-score, false positive rate, precision-recall AUC, detection latency, and computational overhead for the assessment of intrusion detection. The additional criteria are robustness, explainability, and cross-device generalization.

Keywords: IoMT Security, Zero-Day Attacks, Intrusion Detection, Self-Supervised Learning, Graph Neural Networks, Anomaly Detection, Explainable AI.

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Chapter 1

INTRODUCTION

1.1 Background of the Study

With the rapid progress of Internet of Things (IoT) solutions, the medical industry has witnessed the appearance of the Internet of Medical Things (IoMT).

IoMT represents an ecosystem of interconnected medical devices, software applications, communication tools, and healthcare information systems that employ medical information for its collection, exchange, processing, and analysis purposes.

IoMT solutions generate benefits in terms of remote patient monitoring, advanced diagnostics, automated treatments, and other healthcare-specific tasks.

Among the most common devices belonging to the IoT in healthcare domain are the following:

1. Heart-rate wearable sensors
2. Insulin pumps
3. Connected infusion pumps
4. Implantable cardiac monitors
5. Continuous glucose monitoring systems
6. Remote patient monitoring devices
7. Connected medical imagery devices
8. Smart hospital equipment

These devices relay sensitive medical data using various communication technologies.

Such as Wi-Fi, Bluetooth Low Energy (BLE), Zigbee, cellular networks, and healthcare-specific communication tools.

While creating an ecosystem of interconnected medical devices, IoMT solutions introduce an extended attack surface that requires additional security measures.

Unlike traditional IT solutions, IoT medical devices comprise technologies that directly or indirectly impact the health-state of a human being.

As such, healthcare-specific IoT solutions demand specific considerations in terms of cybersecurity.

A successful attack on a medical device may lead to adverse effects for physical health and even cause fatalities.

For instance, an attacker may benefit from disrupting a medical gateway that relays physiological measurements thus impacting the diagnosis and treatment process.

In a similar manner, an attacker may tamper with the settings of an infusion pump which may result in severe conditions for a patient.

Accordingly, IoMT solutions demand additional considerations in terms of cybersecurity that go beyond the traditional confidentiality, integrity, and availability (CIA) framework.

Security Characteristics of IoMT Environments

IoMT (Internet of Medical Things) environments possess several distinctive characteristics that could impact cybersecurity protection. Some examples are as follows:

Device heterogeneity: medical networks employ a variety of devices designed by different manufacturers that operate on various operating systems, communication platforms, firmware versions, and security capabilities.

Resource constraints: many medical devices have limited central processing unit (CPU), memory, storage, and power; insufficient energy supply could impair device operation.

Legacy technology: healthcare institutions account for medical devices that are unable to process firmware upgrades and other cybersecurity provisions.

Constant availability: some medical systems require continuous operation because their power down could disrupt healthcare services.

Sensitive data: IoT medical devices host personal and sensitive information that should be protected by strict confidentiality and integrity policies.

Moreover, IoMT devices involve multiple communication channels and platforms, including gateways, electronic health records, cloud storage, and other interconnected systems and technologies. Therefore, medical devices are vulnerable to cybersecurity risks that could disrupt healthcare services and endanger human lives.

1.2 Overview of Zero-Day Attacks

Zero-day vulnerability refers to a weakness in a software, firmware, hardware, or protocol that is not yet known to the relevant defender or for which there is not yet a timely effective remedy.

A zero-day exploit takes advantage of such a vulnerability.

A zero-day attack occurs when an adversary uses a zero-day exploit against a target.

The distinction is important because an unknown attack pattern does not necessarily represent a zero-day exploit.

For example, if an IDS fails to detect a sequence of network attacks not previously seen, it does not automatically mean that an unknown vulnerability has been exploited.

1.2.1 Characteristics of Zero-Day Attacks

Zero-day attacks pose difficulties because of the following reasons:

- The exploitation vectors employed might not be detectable by known signatures.
- The security administrator may not possess any vulnerability-related protection mechanism.
- The IDS model may not have enough exposure to such malicious activities.
- Detecting such attacks may require detecting behavioral anomalies.
- Malfunctioning of medical devices due to such attacks may manifest as normal behavior.

1.2.2 IoMT Zero-Day Attacks

Some possible impacts of the successful exploitation of a zero day in IoMT could be:

Table 1.1: Healthcare impact

Security impact	**Potential healthcare consequence**
Device unauthorized access	Inaccurate physiological readings
Compromised data integrity	Inaccurate physiological readings
Service disruption	Service disruption
Compromised confidential information	Data leakage
Compromised device	Uncontrolled use of connected equipment
Lateral movement	Compromise of other hospital systems
Compromised firmware	Malicious persistent behavior
Data manipulation	Timing delay or data alteration

All of these examples show how crucial it is to develop detection approaches that can detect suspicious activities regardless of any prior knowledge of particular exploits.

1.3 Problem Statement

Current IDS for IoMT cannot recognize previously unseen attacks. Signature-based IDS relies on the knowledge of attack patterns. supervised machine learning uses labeled examples; and traditional anomaly detection may suffer from many false positives because of behavioral changes in medical devices.

Moreover, evaluation techniques can misrepresent IDS performance if training and test datasets have similar types of attacks. Other problems include lack of interpretability, inadequate investigation of inter-dependencies between devices, and computational difficulties.

The key research question is:

Is there a possibility to develop a generalized, interpretable, and computationally-efficient IDS that would be able to detect previously unseen attacks in heterogeneous IoMT environments with a low level of false positives and adequate detection latency?

The suggested research involves the development of an IDS based on the self-supervised learning and graph anomaly detection approach. The evaluation will differentiate the detection of zero-day attacks and attacks on devices with already-known vulnerabilities.

1.4 Research Motivation

This research is motivated by the growing reliance of healthcare institutions on connected medical technologies and the shortcomings of traditional security measures.

1.4.1 Patient Safety

Connected devices may play an important role in medical procedures.

An attack aimed at changing the device's behavior or interfering with its communication channels or medical data may result in significant safety implications for patients.

An efficient detection system needs to be able to detect potential threats in time to react to them.

1.4.2 Constraints in Traditional IDS

A signature-based system works well when there is knowledge about the attack pattern.

But an attack pattern can also be one which was unknown and hence not covered by the signatures.

This leads to a requirement of security systems that detect behavior patterns instead of relying on attack patterns alone.

1.4.3 Advances in Artificial Intelligence

New advances in the field of self-supervised learning, graph-based learning, and explainable AI present possibilities in detection of network anomalies.

1.4.4 Practical Implementation Challenges

Resource-constrained devices and continuous availability are typical requirements in IoMT environments.

Hence, security mechanisms must have minimum operational overhead and not cause much computation to be performed on medical devices.

The proposed solution will offload complex computations to an edge gateway or a monitoring server instead of performing them directly on medical devices.

1.5 Research Objective

The key objective of the current research is to:

Design, implement, and analyze the performance of an explainable and adaptable framework for intrusion detection based on self-supervised learning and graph-based anomaly detection that is able to detect unknown cyberattack behaviors in IoMT systems.

Such an approach will be tested in zero-day-like experiments with explicit restrictions on assumptions about unknown vulnerability exploitation.

1.6 Research Goals

During this research, the following goals will be achieved:

GO1: Investigation of cybersecurity threats, vulnerabilities and zero-day detection methods in IoMT environment.

GO2: Discovery of limitations of signature-based, supervised, unsupervised and deep learning intrusion detection approaches for previously unseen attacks.

GO3: Development of self-supervised learning approach for learning meaningful representations of normal behavior of IoMT networks.

GO4: Creation of graph-based anomaly detection algorithm to identify abnormal communication relationships among medical devices.

GO5: Integration of explainable AI algorithms that provide meaningful explanations for anomalies.

GO6: Testing of the proposed solution compared to baseline detection algorithms in reproducible zero-day-like experiments.

GO7: Investigation of the performance and suitability of the proposed framework for deployment in edge environments of healthcare sector.

1.7 Research Questions

The research will address the following research questions.

ID: Research question

RQ1: What constraints impact existing intrusion detection methods for detecting novel attacks in IoMT networks?

RQ2: To what extent does self-supervised learning enable the representation of normal communication activities in IoMT environments without large sets of labeled attack data?

RQ3: Is graph representation of communication activities beneficial in detecting novel attack behavior compared to non-graph-based methods of anomaly detection?

RQ4: To what degree can explainable AI generate faithful explanations of abnormal activities within the IoMT network?

RQ5: How well does the proposed approach outperform traditional machine learning and deep learning IDS solutions on held-out attacks?

RQ6: Does the proposed framework allow achieving real-time detection capabilities within edge-based IoMT network monitoring system?

1.8 Research Hypotheses

The following hypotheses will be tested:

H1: Detection Performance

The developed hybrid approach is hypothesized to have a better macro-averaged F1 score on out-of-sample attack classes compared to standard IDS baselines using supervised training.

H2: Graph Representation

Graph-based communication characteristics are hypothesized to yield a much better detection performance on unseen attacks compared to the same architecture but without the graph-based communication representation.

H3: Self-Supervised Learning

Self-supervised representation learning is hypothesized to produce better cross-device generalization compared to fully-supervised models.

H4: False Positive Reduction

A calibrated approach to anomaly detection is hypothesized to have a lower false positive rate than an uncalibrated anomaly detection algorithm at the same detection rate.

H5: Computational Efficiency

The developed framework is hypothesized to show acceptable detection latency and resource requirements on a representative edge computing device. The exact thresholds will be determined before testing based on the experiment setup and the selected monitoring scenario.

1.9 Importance of the Research

1.9.1 Scientific Importance

The research provides additional data for the scientific field of detection of cyberattacks that have never been seen before. Moreover, the research provides the evidence about the level of improvement provided by the relational network information in anomaly detection within heterogeneous IoMT systems.

1.9.2 Technical Importance

The proposed framework aims at improving:

- Novel behavior detection
- Attack generalization
- Device communication relations understanding
- Alerts interpretation
- False positives management
- Feasibility of edge-based detection

1.9.3 Healthcare Importance

The research might provide additional support for security monitoring tools and help to increase visibility of suspicious activity in the medical network. Early detection of attacks would allow security teams to investigate incidents and select adequate reaction.

However, the framework will not cut off safety-critical devices automatically without having a clinical safety policy approved independently.

1.9.4 Academic Importance

The research will provide a set of methods for testing the zero-day-like attacks detection claims. In particular, this is necessary since classification accuracy in traditional datasets does not guarantee attacks on unknown vulnerabilities detection.

1.10 Scope of the Research

The current research explores network intrusion detection systems applied to Internet of Medical Things (IoMT) networks. The major aspects considered within the study include:

1. Monitoring of IoMT devices' network communication.
2. Feature extraction from network traffic data.
3. Self-supervised learning.
4. Anomaly detection based on graphs.
5. Explainable artificial intelligence (XAI).
6. Held-out attacks testing.
7. Edge-deployment feasibility.

The current research focuses on detecting malicious actions rather than developing exploits. Attacks will be tested within the isolated environment using synthetic traffic, publicly available dataset, or specially created vulnerable test systems. There will be no penetration of operational medical devices or accessing patients' systems.

Research Boundaries

The following tasks will not be pursued within the study:

- Discovery of vulnerabilities in commercial medical devices' firmware.
- Exploitation within the healthcare system.
- Detection of any potential zero-day attacks.
- Development of substitute systems for clinical safety.
- Automatic shutdown of IoMT devices.

1.11 Expected Research Contributions

Contributions expected include:

Contribution 1: A combination of self-supervised and graph-based detection approach for protecting IoMT networks.

Contribution 2: A methodological way of evaluating the system through the creation of scenarios where an entire category of attacks is unknown in training.

Contribution 3: Graph-based device-aware representation of IoMT network communication.

Contribution 4: Explainable method of alert generation through the identification of key network attributes and relationships between suspicious devices.

Contribution 5: A reproducible experimental pipeline including pre-processing of data, evaluation of the model, and analysis.

Contribution 6: Feasibility study of edge deployment under realistic computational limitations.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The present chapter endeavors to conduct a literature review of existing research pertaining to cybersecurity of the Internet of Medical Things (IoMT), zero-day attacks, intrusion detection, artificial intelligence, and anomaly detection. The literature review would help establish the theoretical basis for the proposed framework and point out the limitations in current techniques for detection. Six major areas are considered for literature review:

1. Architecture of IoMT and security issues associated with it
2. Zero-day attack detection
3. Machine learning based intrusion detection
4. Deep learning and self-supervised anomaly detection
5. Security analytics using graphs
6. Explainable AI and edge-based detection

A systematic review approach will be followed in conducting the literature review.

2.1 Architecture and Security of IoMT

IoMT networks generally have several layers interconnected to each other.

2.1.1 Layer of the Medical Device

It includes sensors, wearable devices, medical instruments, and patient monitors. These devices may acquire physiological parameters, such as:

- Heart rate
- Blood glucose
- Blood pressure

- Oxygen saturation
- Body temperature
- Status of the device

Security threats may include access to the network, modified firmware, poor authentication, and tampering with sensor data.

2.1.2 Communication Layer

This layer is responsible for the data exchange between the medical devices, gateways and health information systems. Security issues related to this layer are:

- Unauthorized network access
- Eavesdropping
- Communication disruption
- Replay attack
- Abuse of protocols
- Man-in-the-middle attack

2.1.3 Edge and Gateway Layer

This layer aims at consolidating data from devices and may offer local processing and security monitoring. Taking into account the fact that edge gateways have more computing power compared to medical sensors, these gateways can be considered as good places for implementing advanced intrusion detection models.

2.1.4 Cloud and Healthcare Application Layer

Cloud computing and healthcare applications allow storing, processing, visualizing, and accessing data remotely. Threats can include unauthorized access, credential theft, misconfiguration, and information leakage.

2.1.5 Security Considerations

Compromise of one device/gateway might affect others as well since they are interconnected. Thus, security analysis needs to take into account not only individual behavior of the devices but also their interaction in the network as a whole. That is why the study of graph-based intrusion detection seems reasonable.

2.2 IoMT Zero-Day Vulnerabilities

2.2.1 Definition and Classification

A zero-day attack refers to the exploitation of vulnerabilities that the defense party is unaware of or has not found ways to address at the present moment. In IoMT settings, some of the sources of zero-day vulnerabilities can be:

- Firmware of the medical devices
- Embedded operating systems
- Network communication protocols
- Authentication methods
- Device management facilities
- Third-party software components
- Gateway services

Zero-day vulnerabilities could lead to any of the following outcomes:

2.3.2 Detection Challenges

The chief drawback in identifying a zero-day exploit is that there are no attack signs from the earlier occurrences which one can rely on.

Hence, a system detecting attacks should be able to recognize the potentially harmful activities through their side symptoms or indicators.

The indicators or symptoms of such activities can be:

- Device-to-device communication at times which are completely new.
- A sudden change in network traffic quantities.
- Protocol behavior of a device becoming out of ordinary.
- Device communication frequency changing.
- Access to some service at a time which a device normally wouldn' t.
- Device deviating from its assigned responsibilities through its behaviors.

On the other hand, these behaviors may not be restricted to the ones which indicate an attack on a system.

The fact is, activities such as regular firmware updates, system checks, clinical emergencies, device settings, etc., may generate signs or symptoms similar to those of an attack.

This calls on the necessity that any tool which finds anomalies in the system should maintain an optimal balance between high detection power on the one hand and low number or even elimination of false alarms on the other hand.

2.3.3 Zero-Day Detection Versus Unknown-Attack Detection

At this time, I have learned that most of what the paper is proposing is quite focused.

Concept Meanings Required experimental evidence Known attack detection: Detection of those attacks that have already been represented during training; Standard labeled test data

Unseen attack detection: Finding attack types not covered in the training examples; The attack-family held out evaluation

Novel behavior detection: Recognition of behaviors that were never taught or learned (not only through the normal pattern of behavior but also in case of deviation). This can be referred to as Anomaly detection evaluation

Zero-day exploit detection: Recognizing the fact by identifying the activity caused through exploit of a vulnerability that was not known or addressed at that time The source of exploit and chronology will be proved as evidence of such exploitation Zero-day-like evaluation: It is a simulation of limited attack knowledge before the time.

A part of the proposed research will also consist of assessing unseen attack detection and zero-day-like generalization, mainly, I guess.

At best we only can make a zero-day exploitation claim if there comes independently documented evidence with verification, that is, evidence that supports our claim or the proof that we present is from the same source.

2.4 Methods of Detecting Intrusions That are Traditional in Nature

There is a distinction between systems that have characteristics of an attack (signature), and systems that identify intrusions based on deviations from normal activities (anomaly).

2.2.2 Signature-Based IDS

Signature-based systems detect patterns by comparing observed activity to pre-defined attack patterns.

Some common tools that detect based on a set of rules are network-based tools like Snort and Suricata.

Pros:

- Ability to detect familiar attacks.
- Eggs in one basket Rules can be interpreted by anyone reading them.
- Not many layers in alert investigation.

Cons:

- Basing defense on only the presence of the right tool (signature) for the job.
- Low success rate for discovering exploitation techniques that aren' t well known.
- Maintain the right tools (signatures) all the time.

2.2.3 Anomaly-Based IDS

Anomaly detection techniques rely on identifying outliers from standard behavior. For instance.

Common methods include:

- Modeling anomalies via statistics.
- Identification of commonalities via a single group.
- An effective isolation technique in the case of small data samples.
- A classifying technique used in support vector machines that requires very few data samples (one-class).
- Capture of the input features to the network by using an encoder that is part of the network.
- Predicts the future by learning the underlying patterns in some set of inputs over time.

Benefits

- A chance to detect behavior that we have never seen before.
- A lesser reliance on signatures of known attacks.
- Being able to learn normal operations that are different from one computer to another.

Drawbacks

- Wrong alarms or false positives.
- The system gets confused because the users change the behavior in a legitimate way.
- It' s hard to define what normal behavior means when the network is full of various devices working in different ways it' s almost like describing a rainbow.

- There is a certain chance that the system will learn some of the techniques of malware if there is a change in how a machine functions as a result of such techniques (model updating/retaining).

Limitations such as learning malicious behavior during updates call for a more systematic approach in defining behavior.

2.5 Machine Learning-Based Intrusion Detection

Machine learning for intrusion detection has mostly been limited to supervised algorithms, which classify network based on historical samples of attacks.

The main supervised algorithm are:

Random Forest

Random Forest constructs a classifier by combining the output of various individual decision trees.

Its ability to perform well on complex interactions in the presence of a large number of variables and also its excellent handling of network flow data with a mixture of numerical, categorical and textual features have made it one of our go-to baselines.

Support Vector Machine

Support Vector machines try to find the optimal separating hyperplane to minimize classification error. If a set of training data represents the normal behavior, One-Class SVM is used to detect anomalies.

When a dataset is composed mostly of one class, e.g. normal samples, a technique such as novelty detection can be used to automatically find the class.

XGBoost

Gradient boosted tree methods usually outperform traditional tree-based models like random fores t and single decision tree if the trees are properly structured and tuned in the dataset.

Despite all their strengths, most supervised models are trained on data that may represent only a small subset of the different types of attacks.

Isolation Forest

Isolation Forest is very similar to Random Forest.

Besides its simplicity, its biggest advantage is not needing labels or attack categories for the dataset because isolation is purely a property of the data rather than a modeling assumption.

Bottlenecks of Conventional Machine Learning

Major drawbacks are:

- The need for large amounts of training data that capture the full spectrum of the behavior of the system. Model predictions can become unreliable if the model is trained on data whose distribution differs significantly from the test data. Model cannot exploit and communicate relationships between the features well. It can be

very easily overfitted especially to unique dataset features. Inability to generalizing across devices and situations.

Comparing the performance of our hybrid framework against traditional machine learning models is part of the proposed work.

2.3 Deep Learning for IoMT Intrusion Detection

Deep learning models can learn complex relationships from network traffic.

2.3.1 Autoencoders

An autoencoder consists of an encoder and a decoder.

The encoder transforms input features into a lower-dimensional representation.

$$z = f_\theta(x)$$

The decoder reconstructs the original input.

$$\widehat{\mathbf{x}}{=}\mathbf{g}_\phi\left(\mathbf{z}\right)$$

The reconstruction loss can be expressed as:

$$L_{rec} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - \widehat{x}_i\|_2^2$$

An observation with unusually high reconstruction error may be considered anomalous.

However, autoencoders can sometimes reconstruct malicious observations effectively, particularly when attacks resemble normal traffic.

Therefore, reconstruction error alone may not provide sufficient discrimination.

2.6.2 Long Short-Term Memory Networks

LSTMs are the type of recurrent networks that are best at modeling the relationship of the present with the past or the past with the present.

For network traffic sequences, one may use these networks to detect outliers from the regular temporal pattern of a process.

For example, LSTM-based system might be developed to identify message frequencies that are out of the norm or identify normal intervals.

However, LSTM-based systems may demand a quite large training dataset and powerful hardware.

These are typical situations where the model can take hours of training with high-performance systems which are not always accessible at the local level of the edge systems.

2.3.2 Transformer-Based Models

Transformers mainly rely on an attention mechanism to model the relations among the data points in an observation sequence.

Thus, they can potentially learn very complex dynamics in the temporal features of the input data like the network traffics.

However, transformers are not a suitable alternative when it comes to the constrained edge systems because they are computationally demanding in nature.

2.3.3 Research Implications

DL can extract high-level temporal features, but it is model complexity does not lead the generalization towards unseen attack patterns automatically. To prevent from the possible pitfalls of DL, a cautious study of performance, coupled with thorough benchmarks with other approaches is essential.

2.4 Self-Supervised Learning for Zero-Day-Like Detection

SLL techniques allow the training of a system using large volumes of unlabeled data, with no external expert intervention, as the only supervisory signal in these techniques is an artificial label generated from the data itself.

That is, rather than being told what each label should be, one is given information about the labels that must be derived from the unlabeled data only. For example, one might learn a mapping between images or text, without any human annotations, simply by the fact that a particular image or text is the translation of another.

In the case of cyber threat detection, especially of IoMT environment, having fully manual annotated examples is not practical.

2.4.1 Contrastive Learning

Contrastive learning aims to produce similar representations for observations that are related while ensuring that those that are not related are separated.

For instance, two valid augmentations of the same network-flow sequence should be considered as positive pairs.

A contrastive objective can be expressed using the InfoNCE loss:

$$L_{SSL} = -\log \frac{\exp\left(sim\left(z_i, z_i^+\right)/\tau\right)\right)}{\sum_j \exp\left(sim\left(z_i, z_j\right)/\tau\right)\right)}$$

Where:

- z_i is the representation of the original observation.
- z_i-plus: the representation of a positive sample.
- \(sim\): Similarity function.
- \(\tau\): Temperature parameter.

When it comes to IoMT data, the augmentations have to ensure that security-relevant information is preserved.

Altering at random the identities of devices, the states of the protocols, or the important timing features might result in unrealistic observations.

2.4.2 Masked Feature Modeling

A different method involves hiding certain input features and then training the model to reconstruct those features.

Instead, the model memorizes the attack labels rather than learning the relationships between network features.

2.4.3 Advantages

Self-supervised learning may provide:

- There will be less reliance on attack labels.
- Improved feature representation.
- The ability to handle diverse types of traffic.
- It might result in better transfer between devices.

2.4.4 Limitations

Self-supervised learning does not by itself guarantee zero-day detection.

The way it performs is determined by the quality of the training data, the representation objectives, and the relationship between the learned features and malicious behavior.

The research that has been proposed will carry out an experimental investigation into these limitations.

2.5 Graph-Based Anomaly Detection

Methods based on graphs represent the entities of a network and the relationships between them.

An IoMT network can be represented as:

$$G = (V, E)$$

Where:

- V: A set of nodes that represent devices or services.
- E: The set of communication relationships.

For example, nodes may represent:

- Patient monitoring devices.
- Medical gateways.
- Healthcare servers.
- Workstations.
- Cloud services.

Edges can represent the communication sessions between these entities.

2.8.1 Graph Neural Networks

Graph Neural Networks (GNNs) create representations by gathering information from nodes.

A standard message-passing approach is:

$$h_v^{(l+1)} = \sigma\left(W_l h_v^{(l)} + \sum_{u \in N(v)} \alpha_{vu} W_n h_u^{(l)}\right)$$

Where:

- $h_v^{(l)}$: Node representation, at layer l.
- $N(v)$: Neighboring nodes.
- W_l, W_n: weight matrices.
- α_{vu} :Neighbor contribution weight.
- σ:Activation function.

2.5.1 Graph Attention Networks

Graph Attention Networks give weights to the nodes that are connected. This helps to focus on the important connections. For example, in a hospital network communication between a monitor and its authorized gateway is expected. If the monitor starts talking to a workstation that is not related that may be a sign of something wrong. By using attention, the model can learn which connections matter more.

2.5.2 Dynamic Graphs

In IoMT the way devices connect can change over time. To capture this dynamic graphs use time- edges. At each time point the graph is defined as:

$$G_t = (V_t, E_t)$$

This allows the system to track changes in how devices communicate. A temporal graph model can explore:

- connections that appear between devices.
- Unusual interactions that were not seen before.
- Changes, in how devices communicate.
- Strange traffic patterns that happen over time.

2.5.3 Research Gap

Graph-based techniques offer a way to find unusual relationships.

It is important to check how useful these techniques are, for finding new types of problems in different kinds of IoMT environments.

The research will check if using graph information gives advantages compared to older ways of looking at data flows.

2.6 Explainable Artificial Intelligence

Explainable Artificial Intelligence or Explainable AI aims to make the decisions made by machine learning models easier to understand. This is especially important in healthcare cybersecurity. Security analysts often need to explain why an alert was triggered before they take any action that could impact care or clinical operations. If a system raises a warning the analyst must be able to justify it quickly.

2.6.1 SHAP

SHapley Additive exPlanations, known as SHAP is a method used to explain how each feature in a model contributes to a prediction. It helps to break down the prediction and shows which parts of the data had the influence. In the context of network security SHAP can be used to understand what caused an anomaly score. For example it can show whether the score was driven by packet frequency or an unexpected destination or a long session duration. Some of the features that SHAP can help explain include:

- Unusual packet frequency
- Unexpected destination
- Abnormal session duration
- Increased communication volume
- Changes in protocol usage

2.6.2 LIME

Local Interpretable Model-Agnostic Explanations or LIME provides explanations for predictions made by any model by approximating it with a interpretable model around a single prediction. This method is useful when you want to understand why a specific network event was flagged as suspicious. It gives an explanation meaning it focuses on one instance at a time. This makes it helpful for analyzing network observations and understanding what the model found unusual, about them.

2.6.3 Graph Explanations

Graph-specific explainability methods can point out which nodes, edges or parts of the graph are most important. For example an explanation might show a connection, between a medical device and a service it shouldn't be talking to. This helps spot patterns that could mean something bad is going on.

2.6.4 Limitations of Explainability

Just because an explanation looks right doesn't mean it's actually correct. A model might say certain features are important because they often show up with attacks. Not because they're the real cause. That's why the research will check how accurate and consistent the explanations are. It won't take for granted that something that looks clear is also trustworthy.

2.7 Representative Research and Foundational Frameworks

The works listed below lay the groundwork for the proposed research.

Table 2.1: Representative Research and Foundational Frameworks

Study or framework	Main contribution	Relevance to the proposed research
Garcia et al (2009)	Survey of anomaly-based network intrusion detection	Establishes foundational IDS concepts
Ruff et al (2021)	Comprehensive review of deep and shallow anomaly detection	Supports anomaly detection methodology
Chen et al (2020), SimCLR	Self-supervised contrastive representation learning	Provides an SSL foundation
Velickovic et al. (2018), GAT	Graph attention mechanisms	Supports relational feature learning
Kipf and Welling (2017), GCN	Graph convolutional learning	Provides a graph modeling baseline
Lundberg and Lee (2017), SHAP	Model prediction explanations	Supports interpretable anomaly alerts
FDA (2023)	Cybersecurity guidance for medical-device submissions	Establishes medical-device cybersecurity context
MITRE ATT&CK	Structured representation of adversary tactics and techniques	Supports threat modeling
NIST SP 800-30	Information security risk assessment guidance	Supports systematic threat and risk analysis

Important literature limitation: These sources are foundational and regulatory but they do not provide a verified exhaustive review of the newest IoMT zero-day detection papers. The proposed research must also find, check and critically compare peer-reviewed IoMT studies published through 2026.

The proposed research will therefore conduct a literature review before finalizing the proposed research claims of novelty

2.7.1 Literature Review Search Strategy

The literature search will use databases such, as:

- IEEE Xplore
- ACM Digital Library
- ScienceDirect
- SpringerLink
- PubMed

Here' s a sample search phrase you might try:

- ("Internet of Medical Things" OR IoMT OR "medical IoT")
- ("zero-day" OR "unknown attack" OR "unseen attack" OR "novel attack" OR "anomaly detection")
- ("intrusion detection" OR "self-supervised learning" OR "graph neural network" OR "deep learning")

Inclusion Criteria
We will include studies that meet these points:

1. They focus on the Internet of Medical Things (IoMT), medical IoT, or related cyber-physical systems.
2. They deal with detecting intrusions, spotting anomalies, or identifying unknown attacks.
3. They clearly explain a method used to detect issues.
4. They provide measurable results from evaluations.
5. They are peer-reviewed research papers or official technical standards.

Exclusion Criteria
We will leave out studies if they:

1. Don't give enough details about their methods.
2. Don't include any proper security assessments.
3. Are just repeats of research we' ve already considered.

4. Make bold claims about zero-day detection without backing them up.

5. Don't focus on network or device behavior security at all.

I notice that the main review period covers 2020 to 2026 and foundational studies supplement it.

I also see that a PRISMA-inspired screening and reporting process will document identification, screening, eligibility and inclusion decisions.

2.8 Research Gaps

Based on the limitations found in the literature the research will look into the following possible gaps.

These gaps will be. Adjusted after finishing the systematic review.

Gap 1: Limited Evaluation of Unseen Attacks

Many intrusion detection evaluations use random dataset splits.

These methods can put examples from the attack types in both training and testing sets.

Because of that the accuracy reported might not show how well the system works against attack behaviors.

Proposed response: Use held-out attack-family evaluations.

Gap 2: Dependence on Labeled Attack Data

Supervised methods need labeled examples.

Such labels might not exist for ways of exploiting systems.

Proposed response: Explore self-supervised learning using barely labeled traffic.

Gap 3: Not Enough Relational Context

Flow-based classification can miss interactions between devices.

Proposed response: Add graph-based ways to show how devices communicate.

Gap 4: False-Positive Alerts

Anomaly detection can mix up changes, in operations with actual attacks.

Proposed response: Add validation-based threshold calibration and device-aware anomaly scoring.

Gap 5: Not Enough Explanation

Deep learning models can create alerts without reasons.

Proposed response: Add explanation tools at the feature level and the graph level.

Gap 6: Deployment Challenges

Some models need computing power than medical devices can offer.

Proposed response: Check how well the models work at the edge gateway and compare the model size with the detection benefits.

2.9 Conceptual Research Framework

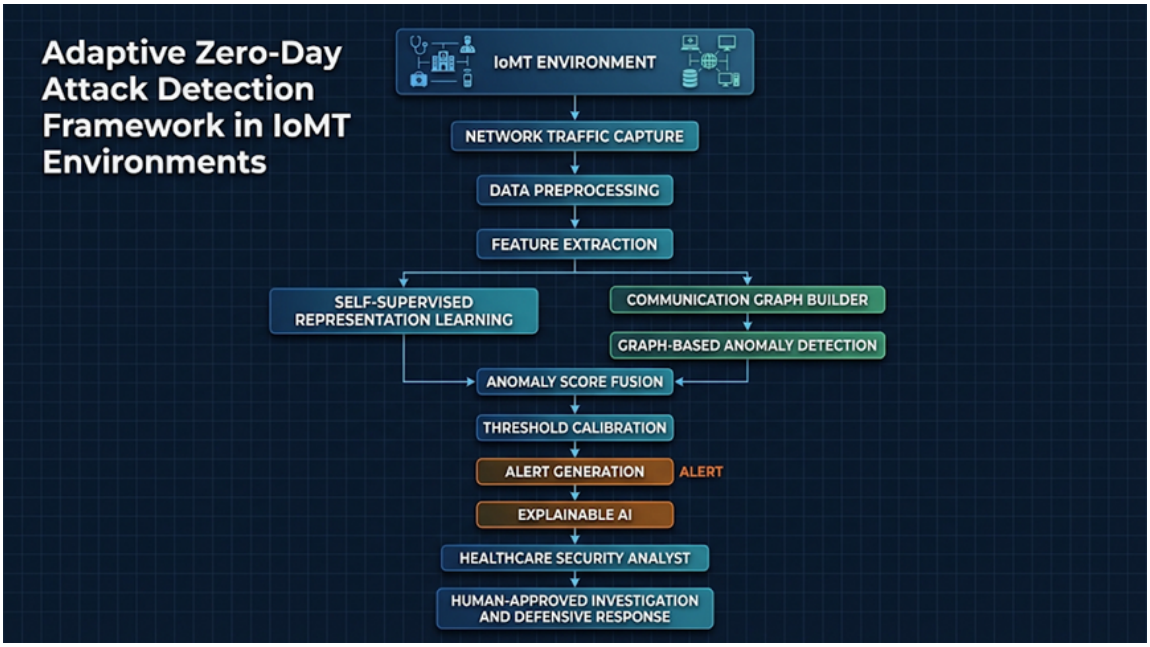


Figure 2.1: Research Framework

This architecture forms the basis of the research methodology.

Chapter 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter shows the research design, the steps the ways data will be collected the detection framework we propose the metrics we will use for evaluation and how we will make the work reproducible.

The study will use a experimental research approach.

The main goal is to find out whether combining self-supervised representation learning with graph-based anomaly detection makes it easier to spot attack behaviors in IoMT networks.

The methodology will emphasize:

- Controlled experiments.
- Reproducibility.
- Fair comparisons with methods.
- Clear separation of training and testing attack categories.
- Tests for statistical significance.

- Feasibility of real-world deployment.

Self-supervised representation learning and graph-based anomaly detection are central, to this study.

3.2 Research Design

The research will use a design science and experimental evaluation method.

Design science will help create the intrusion detection framework that is being proposed.

Experimental evaluation will check if the framework that is being proposed offers improvements compared to current methods.

The research will be split into seven phases.

Table 3.1: Phases for Research

Phase	Action	Expected Output
1	Literature review and threat modeling	Research requirements and threat modeling
2	Dataset collection and simulation design	Experiment results
3	Feature engineering and data processing	Well-structured, clean data sets
4	Self-supervised model creation	Behavioral Models
5	Graph-based outlier detection	Relational anomaly scores
6	Explain ability and model integration	Full detection framework
7	Assessment and statistical analysis	Research results

3.3 Threat Modeling

Detection of assets, attack surfaces, adversary goals, and potential security effects through threat modeling.

3.3.1 Protected Assets

The protected assets are:

- Medical device availability.
- Accuracy/precision in medical measurements.
- The way the device communicates.
- Patient monitoring services.
- The health gateway infrastructure.
- The setup of the medical network.
- Secure health records.

3.3.2 Potential Adversaries

The threat model will identify the following:

External attackers: Individuals seeking to obtain unauthorized access by means of open up-to-exposed network services.

Compromised internal devices: Devices which have been hacked and show unauthorized network activities.

Malicious insiders: Privileged individuals who exploit their access rights improperly.

Supply-chain threats: Hacked firmware or software components being added via third-party dependencies.

3.3.3 Threat Categories

Behavioral signs that the research will study include:

- Unauthorized access. Port scanning/Network sweeping. Denial-of-service (DoS/DDoS) attacks. Exfiltration of data. Lateral mobility. Command-and-control (C2) communications. Protocol abuse/tampering. Abnormal behavior of the device.

These categories only show what can potentially be observed while they do not confirm on their own that it was a case of exploitation through an unknown (zero-day) vulnerability.

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3.3.2 STRIDE Threat Modeling

We will use the STRIDE framework to classify threats.

Table 3.2: STRIDE Categories and IoMT Examples

STRIDE Category	**IoMT Example**
Spoofing	Unauthorized entity impersonating a medical device.
Tampering	Manipulation of transmitted monitoring data.
Repudiation	Lack of reliable audit evidence of unauthorized activity.
Information Disclosure	Disclosure of confidential patient information.
Denial of Service	Interference with communication between medical devices.
Elevation of Privilege	Unauthorized acquisition of elevated device privileges.

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3.3.3 Outcomes of Threat Modeling

The threat modeling phase will deliver: - Inventory of assets.

- Trusted boundary diagram.
- Threat scenarios
- Attack surface analysis etc.
- Risk assessment matrix .
- Detection requirements;

Where appropriate, threat scenarios will be mapped to relevant adversary behaviors using MITER ATT&CK.

3.4 Data Collection

The work will utilize both publicly available and controlled experimental data.

3.4.1 Public IoMT Data Sets

Candidate dataset: CICIoMT2024.
The study will be checked before inclusion: - The dataset's official source and documentation.

- Types of devices available.
- Types of attacks
- Definitions of features.
- Availability of raw packet capture.
- Distribution of classes
- Sample collection.
- Licensing and terms of use of data.

More IoMT or IoT datasets could be considered if they provide relevant attack diversity.
However, datasets from different environments will not be indiscriminately combined, as their feature distributions and collection procedures can be substantially different.

3.4.2 Selection Criteria for Datasets

The selected datasets should preferably contain:

1. Legitimate network traffic.
2. Several classes of attack.
3. Device identifiers or relationships of communication.
4. Information on network flow.
5. Time stamp information.
6. Adequate samples for controlled evaluation.

If the dataset does not provide reliable device identities or communication endpoints, it may not be possible to reconstruct graphs.

In that case, it will be used only for flow-based evaluation, and graph based experiments will be performed on suitable data.

3.4.3 Controlled Simulation

A laboratory environment will be set up to reproduce the representative IoMT communication patterns.

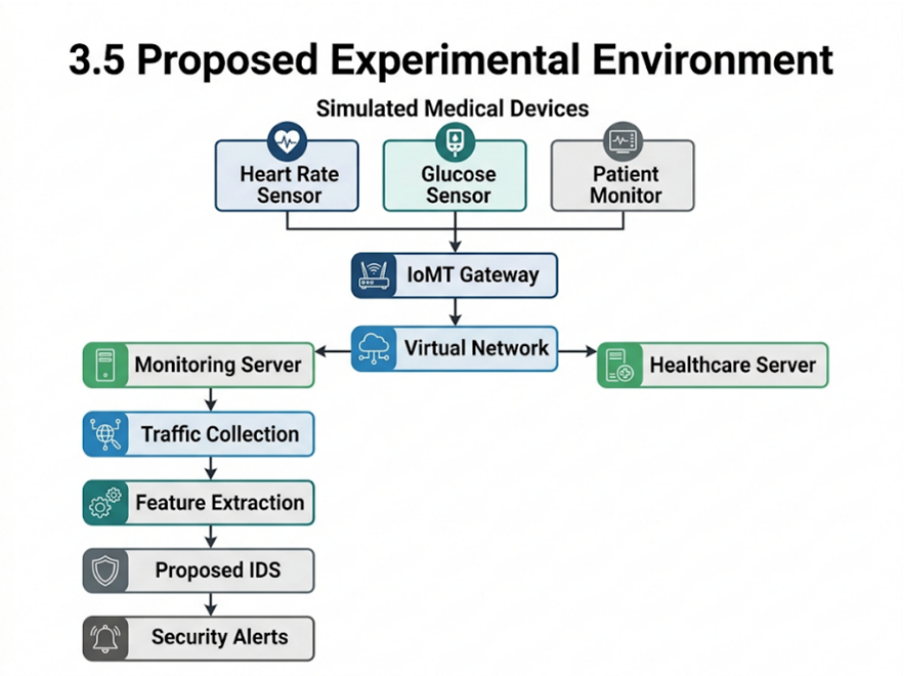
The environment may contain:

- Medical sensor simulation IoMT gateways.
- Application servers for healthcare.
- A network monitoring module.
- A controllable adversarial traffic generator

Only synthetic patient measurements are used.

There will be no real clinical equipment in the simulation.

3.5 Proposed Experimental Setup



The experimental environment will be composed of several logically separated components.

Fig(3.1) experimental environment

“‘latex **Proposed Tools**

Table 3.3: Proposed Tools and Their Purposes

Tool	**Purpose**
Wireshark	Inspecting packet details and verifying network traffic.
Zeek	Monitoring network traffic and generating structured records.
Suricata	Identifying known attack signatures as a baseline IDS method.
Python	Extracting useful data and developing experiments.
Scikit-learn	Implementing traditional machine learning models as baselines.
PyTorch	Developing deep learning models.
PyTorch Geometric	Developing graph neural networks.
NetworkX	Representing network connections and analyzing their structure and behavior.

Tool	Purpose
Docker	Creating a reproducible and deployable experimental environment.
Mininet	Simulating network topologies when needed.
MLflow	Tracking experiments and recording model performance.
SHAP	Explaining model predictions at the feature level.
GNNExplainer	Explaining graph neural network predictions by identifying important graph components.
Pandas and NumPy	Preparing and processing experimental data.
Matplotlib	Visualizing experimental results.

The final selection of tools will depend on the available hardware and the characteristics of the dataset.

3.6 Data Preprocessing

Data Preprocessing will change raw network observations into inputs that a model can use.

Step 1: Data Cleaning

The dataset will be inspected for:

- Missing values.
- Records.
- Invalid feature values.
- Infinite values.
- Inconsistent timestamps.
- Malformed observations.

Duplicate removal will be applied carefully because repeated observations may represent periodic device communication.

Step 2: Feature Selection

Candidate features include:

Table 3.4: Feature Selection

Feature	**Description**
Flow Duration	The total time elapsed during a network communication flow.
Packet Count	The total number of packets transmitted within a flow.
Byte Count	The total number of bytes transmitted within a flow.
Packet Size Statistics	Statistical measures describing the distribution of packet sizes within a flow.
Inter-arrival Time	The time interval between consecutive packets.
Protocol	The communication protocol used for data transmission.
Source Device	The device that initiates the communication.
Destination Device	The device that receives the communication.
Connection Frequency	The number of connections established within a specified time window.
Traffic Rate	The volume of network traffic transmitted per unit of time.
Communication Novelty	An indicator of whether communication between two devices represents a previously unseen or unusual relationship.

Identifiers will be handled carefully to prevent models from memorizing dataset-specific labels.

Step 3: Normalization

Numerical features may be standardized using:

$$x^{'} = \frac{x - \mu}{\sigma}$$

Where?

We have:

- x: the original variable
- μ: the training set mean

- σ: the training set standard deviation

Normalization parameters will be derived from the training partition only.

Step 4: Temporal Segmentation

The network data shall be split into a bunch of time windows.

For example:

$$W_t = \{x_i \mid t \ \leq timestamp_i < t + \Delta t\}$$

Where Δt denotes the chosen length

Many different window sizes

Multiple window sizes are going to be looked into so that can find the right balance between getting the fastest detection and keeping adequate contextual information.

Step 5: Dataset Partitioning

The data will be split as:

- training set.
- validation set.
- and testing set.

If the dataset can support such features, we will perform temporal separation and device-level separation separately.

Only the training data will be considered for the preprocessing stage in order to eliminate the risk of data leakage.

3.7 Proposed Detection Framework

I will describe the four parts of the framework.

- Self-supervised behavioral representation learning.
- Graph-based anomaly detection.
- Anomaly score fusion.
- Explainable alert generation.

3.7.1 Component One:Self-Supervised Representation Learning

Self-supervised representation learning will learn representations of network behavior.

Let: $X = \{x_1, x_2, \ldots, x_n\}$

Here is the input, from the user:

show the network observations.

An encoder changes each observation into a hidden representation:

$$z_i = f_{0(x_i)}$$

The model will be trained using a self-supervised objective.

I think a masked reconstruction objective is an initial choice, for a self-supervised objective:

$$L_{SSL} = \frac{1}{|M|} \sum_{j \in M} (x_j - \widehat{x}_j)^2$$

Locations:

- $\mathcal{M}$ Indicates the indices of the masked features.
- x_jThe observed value of the first feature at the \$j\$th position.
- $\hat{x}_j$: the prediction or reconstruction.

For categorical features, the best reconstruction loss should be selected from a set of available ones, i.e., categorical reconstruction loss.

In case of detecting anomalies, we will work with representations discovered during a training stage.

3.7.2 Scoring Behavioral Anomaly

The behavioral anomaly score can be the only one or among others that is determined on the basis of reconstruction error:

$$s_{\text{behavior}}(x_i) = \|x_i - \hat{x}_i\|_2^2$$

Higher scores mean a difference, from what the system has learned.

A high reconstruction error doesn't prove something is malicious.

It just suggests behavior that stands out not that it's definitely harmful.

3.8 Graph Construction

The second part will show how devices communicate with each other using graphs.

Let:

$G_t = (V_t, E_t)$

Where:

- V_t: Devices that are seen during time window t.
- E_t : Communication links that happen during that window.

Node Features

It is possible to define node features as follows:

- Device type.
- Number of connections.
- Average data transfer rate.
- Unique destination count.
- Protocol type.
- Frequency of interaction.

Edge Features

Edge features can cover:

- Type of connection between source and destination devices.
- Flow count.
- Total volume of data transferred.
- What is the duration of connection.
- Frequency of interaction.
- Type of protocols used.

Graph Creation Steps

Upon each time frame passing:

- Establish which devices communicate.
- Create nodes for observed devices.

- Add edges for the communication connection.
- Collect data related to specific flows.
- Generate features for nodes and edges.
- Save the final graph constructed.

3.9 Graph-Based Anomaly Detection

Graph-Based Anomaly Detection employs a graph network to learn normal relationships among IoMT entities.

Graph-Based Anomaly Detection considers a Graph Attention Network, as a candidate model.

The node representation can be updated using:

$$h_i' = \sigma\left(\sum_{j \in N(i)} \alpha_{ij} W h_j\right)$$

Where:

- $'_i$ \): Updated representation of node i.
- $n(i)$: Neighboring nodes
- α_{ij} : Attention coefficient.
- w : transformation matrix.

Graph-Based Anomaly Detection trains the graph model using a normal-behavior objective.

One possible method is edge reconstruction.

Let:

$$\hat{a}_{ij} = \sigma\left(h_i^T h_j\right)$$

Where:

- $\hat{a_{ij}}$ This is the predicted probability that two nodes are connected in a communication network.
- $_i$, $_j$ These are the learned node representations.

The graph reconstruction loss may be:

$$L_{\text{graph}} = -\sum_{i,j} \left[a_{ij} \log(\hat{a}_{ij}) + (1 - a_{ij}) \log(1 - \hat{a}_{ij})\right]$$

In practice negative-edge sampling and class weighting will be used when needed.

The resulting graph anomaly score will show communication patterns or unusual structural behavior.

3.10 Hybrid Anomaly Score Fusion

The framework will combine anomaly scores and graph-based anomaly scores.

The combined score is:

$$S_{\text{final}} = \lambda S^*_{\text{behavior}} + (1 - \lambda) S^*_{\text{graph}}$$

Where:

- S^*_{behavior}: Calibrated behavioral anomaly score.
- $\left(S^*_{graph} \right)$Calibrated graph anomaly score.
- (λ)Fusion weight between 0 and 1.

We must calibrate both scores to scales before fusion.

We will select the fusion weight using validation data than test data.

An alert will be generated when:

$$[\ S_{final} > \tau]$$

Where τ is the anomaly detection threshold.

Threshold Selection

We may select the threshold using a percentile of normal validation scores:

$$\boldsymbol{\tau = q_{1-\alpha}\left(s_{\text{normal}}\right)}$$

Where α represents the targeted false-positive proportion, on the calibration set.

We will measure the false-positive rate independently on the test data.

Because operational conditions may change we will also investigate threshold stability under distribution shifts.

3.11 Explainable Alert Generation

The framework will give explanations for any anomalies it finds. I think this is helpful.

An alert may contain:

- Alert identifier: A reference for the event.
- Affected device: The device linked to the activity.
- Anomaly score: The score, from the model.
- Observed deviation: A description of the behavior.
- Influential features: The features that contribute to the alert.
- Graph context: The communication relationships that are relevant.
- Confidence information: Alert information that has been calibrated and is statistically justified.

3.12 Zero-Day-Like Experimental Evaluation

This part is one of the important methodological contributions.

A simple random split of training and testing data does not work well for checking how well a system can handle attack types that it has never seen before.

So the study will use different experimental setups.

Experiment 1: Known-Attack Detection

Purpose: Show how a standard IDS works.

The training set will hold traffic and a few known attack kinds.

The test set will hold data that still belongs to those same attack kinds.

This test will give a baseline. It will not prove that the system can find zero-day attacks.

Experiment 2: Leave-One-Attack-Family-Out

Purpose: Check how well the system can spot attack families that it has never seen before.

Assume the data set has attack families:

A={A_1,A_2,A_3,A_4,A_5}

For one experimental run:

A_train={A_1,A_2,A_3,A_4\}

A_{test}={A_5}

The held-out family will be excluded from model training hyperparameter tuning, threshold selection and any attack-specific representation learning.

The process will be repeated for each attack family.

For the normal-only anomaly detector, model fitting will use benign training observations; the held-out-family design will determine the malicious evaluation sets.

Experiment 3: Cross-Device Generalization

Purpose: Evaluate whether the model generalizes to unfamiliar devices.

The framework will be trained using traffic from selected device groups.

Testing will involve device groups excluded from training, where dataset coverage permits.

Experiment 4: Distribution Shift

Purpose: Evaluate robustness under changing network conditions.

Potential changes include:

- traffic volumes.
- Modified communication intervals.
- Additional legitimate services.
- New benign communication relationships.
- Changes, in device population.

Experiment 5: Edge Deployment

Purpose: Evaluate feasibility.

The trained model will be deployed on an edge computing platform or constrained virtual machine.The experiment will measure:

- Inference latency.
- Memory consumption.
- CPU utilization.
- Throughput.
- Model size.

3.13 Baseline Models

I will compare the proposed framework, with baseline models.

Table (3.4) compare the proposed framework

Table 3.5: Baseline Models and Evaluation Purposes

Model	**Category**	**Evaluation Purpose**
Random Forest	Supervised machine learning	Conventional classification baseline.
XGBoost	Supervised machine learning	Strong tabular classification baseline.
One-Class SVM	Novelty detection	Baseline for modeling normal behavior.
Isolation Forest	Anomaly detection	Lightweight anomaly detection baseline.
Autoencoder	Deep learning	Reconstruction-based anomaly detection.
LSTM Autoencoder	Sequential deep learning	Temporal anomaly detection.
Graph-only Model	Graph anomaly detection	Measures the benefits of relational modeling.
SSL-only Model	Self-supervised anomaly detection	Measures the benefits of representation learning.
Proposed Hybrid Framework	Self-supervised learning and graph anomaly detection	Evaluates the proposed approach.

3.14 Evaluation Metrics

Evaluation Metrics will guide the evaluation of the framework, with performance measures.

3.14.1 Confusion Matrix

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Table 3.6: Confusion Matrix for Attack Detection

Actual Class	**Predicted Attack**	**Predicted Normal**
Actual Attack	True Positive (TP)	False Negative (FN)
Actual Normal	False Positive (FP)	True Negative (TN)

Accuracy indicates how many correct predictions have been made. This metric may be deceptive if there is an imbalance between normal and malicious observations.

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision indicates what percentage of the generated alert about possible attacks corresponds to actual attacks**.**

Recall

Measures the proportion of actual attacks that are recalled.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall refers to the measure of the proportion of actual attacks that are recalled.

F1-Score

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1-score: A measure that combines precision and recall.

False-Positive Rate

$$\text{FPR} = \frac{FP}{FP + TN}$$

False positive rate is also critical as too many alerts can be overwhelming to health care security analysts.

Detection Rate

$$\text{DR} = \frac{TP}{TP + FN}$$

The detection rate is the same as the recall of the attack class

Additional Metrics

The study will also report:

- Area under the ROC curve. Area under the precision–recall curve.
- Per-attack-family recall.

- Macro-averaged F1-score.
- False alerts / device hour.
- Detection latency.
- CPU utilization.
- Memory consumption.
- Model inference throughput.

The focus of the paper will be on precision-recall analyses for imbalanced attack detection.

3.15 Statistical Analysis

The research shall implement appropriate statistical analysis to establish if these improvement means anything.

Repeated Experiments

We evaluate models across multiple random seeds and holdout attack-family configurations.

Confidence Intervals

Confidence Intervals will be estimated using appropriately applicable resampling techniques for the given dataset. For datasets with temporal correlation, we will use either block-level or device level resampling as opposed to independent flowlevel sampling.

Statistical Significance

Suitable tests for paired comparisons will be performed (e.g., Wilcoxon signed-rank test; paired permutation tests).

Bootstrap confidence intervals for performance differences.

The statistical tests are chosen depending on the number and independence of experimental units.

Effect Size

These results are reported as differences in performance with associated effect sizes instead of focusing solely on statistical significance.

3.16 Explain ability Evaluation

Explain ability component will be evaluated based on 3 dimensions.

Fidelity

Does this explanation match what we know about how the model makes its decisions?

Stability

Are there reasonably consistent explanations for such similar observations?

Usefulness

Can this help an analyst to better understand what behaviors caused them to trigger an alert? Where possible, we'll try to compare our results with known synthetic attack characteristics. We'll try to not conflate feature importance with evidence of exploit causality.

3.17 Reproducibility

The reproducibility will be an essential element of our proposal for this research. We propose to implement these measures below.

3.17.1 Dataset Documentation

The research will document:

- Dataset source
- Dataset version
- Data collection characteristics
- Preprocessing procedures.
- Included/excluded features.
- Train/validation/test partitions.

3.17.2 Source Code

The experimental code will include modules dedicated to:

$$h_i' = \sigma\left(\sum_{j \in N(i)} \alpha_{ij} W h_j\right) \tag{3.1}$$

3.17.3 Experimental Configuration

Each experiment will record:

Random seed. Model architecture. Hyperparameters.

Training Duration.

Dataset Partition.

Evaluation Metrics.

Software versions. Hardware configuration.

3.17.4 Leakage Prevention

Particular attention is being paid to prevention of data leaks.

For example:

- No test observations are used to fit normalization. No held-out attack samples are used to choose thresholds. No test labels are used while developing models.
- Related traffic windows will not be randomly distributed to partitions where such distributions may introduce dependency.
- Duplicate (or nearly duplicate) observations will be investigated across all partitions

3.18 Ethical and Safety Considerations

We will conduct all of our research following responsible cybersecurity research practices.

Our experiments will only use authorized datasets within isolated environments. We will not conduct any attacks on operational healthcare infrastructure.

Synthetic patient data will be used where necessary for clinical information.

If there are to be any real healthcare data involved at all they would need proper authorization and appropriate levels of privacy protection. Such usage also needs to have been approved by our relevant institutions' ethics committees.

My proposal is that this would work best as a ”monitoring and decision support” mechanism.

A need for human review to consider clinical safety due to defensive responses that impact medical device connectivity.

3.19 Expected Results

Research aims at determining whether the proposed hybrid framework yields any advantages as compared with conventional IDS techniques.

Potential outcomes include:

- Improved detection of held-out attack categories.
- Better representation of normal IoMT communication patterns.
- More informative detection of abnormal device relationships.
- reduced false-positive rates via calibrated anomaly thresholds,
- interpretable alerts to provide meaningful insights for security analysts, and
- evidence regarding the computational feasibility of enabling such systems at the edge

While all these results are just hypotheses for now (and thus possible advantages from conducting such research), there might be some cases where more primitive anomaly detection techniques can achieve comparable performance.

If we found this out, I believe it could still provide interesting insights about the actual benefit and potential drawbacks of adding more complexity into our models.

3.21 Research Limitations

Limitations of this study might be as follows:

Dataset Limitations

Publicly available data sets might fail to capture the heterogeneity across different operating health care networks

Zero-Day Ground Truth

Previously Unseen Attack Categories Are Not Equivalent To Verified Zero-Day Exploits.

Simulation Realism

However, synthetic IoMT traffic might not have covered all aspects of clinical device behaviors.

Generalization

When you train a model on some data, it often performs poorly when applied to an alternative environment.

Computational Complexity

Graph-based Processing Could Increase Memory and Inference Requirements

Explain ability

Limitations include the inability of feature attribution and graph explanations to give definitive causative explanations for malicious behavior, which are explicitly described in the interpretation of these results.

PROPOSED RESEARCH TIMELINE

I am proposing this plan for a research schedule lasting one year (12 months).

Table (3.5) RESEARCH TIMELINE

Table 3.7: Research Timeline and Activities

Month	**Research Activity**
1	Define the research problem and objectives.
2	Conduct a systematic literature review.
3	Complete threat modeling and research design.
4	Acquire datasets and develop the simulation environment.
5	Perform data preprocessing and exploratory analysis.
6	Develop the self-supervised learning model.
7	Develop the graph-based anomaly detection model.
8	Integrate the hybrid detection framework.
9	Conduct known-attack and unseen-attack experiments.
10	Perform ablation, robustness, and edge evaluation.
11	Conduct statistical analysis and interpret the findings.
12	Complete thesis writing, revision, and defense preparation.

PRELIMINARY REFERENCES

These are some examples to help you get started with your research. Before submitting formally check bibliographic details from the actual published sources.

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